

Netflix Data Analysis Using Python

Project Report

Author

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Abstract

Data analysis is an essential process for extracting meaningful insights from raw data. This project performs Exploratory Data Analysis (EDA) on the Netflix Movies and TV Shows dataset using Python. The analysis includes data cleaning, preprocessing, handling missing values, removing duplicate records, and visualizing important patterns using various statistical graphs.

The project demonstrates how Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn can be used to transform raw data into meaningful business insights. The final output helps understand Netflix's content distribution, release trends, ratings, and country-wise production.

Objectives

The primary objectives of this project are:

- Load and understand the Netflix dataset.
 - Perform data cleaning and preprocessing.
 - Handle missing values efficiently.
 - Remove duplicate records.
 - Perform Exploratory Data Analysis (EDA).
 - Create professional visualizations.
 - Generate meaningful insights from the data.
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Technologies Used

Technology	Purpose
Python	Programming Language
Pandas	Data Manipulation
NumPy	Numerical Computation
Matplotlib	Data Visualization
Seaborn	Statistical Visualization

Technology	Purpose
Jupyter Notebook	Development Environment
GitHub	Version Control & Project Hosting

Dataset Description

The project uses the **Netflix Movies and TV Shows Dataset**, which contains information related to Netflix content.

Important Features

- Show ID
- Type (Movie / TV Show)
- Title
- Director
- Cast
- Country
- Date Added
- Release Year
- Rating
- Duration
- Genre
- Description

The dataset provides useful information for analyzing Netflix's content library and identifying trends across different countries and years.

Methodology

The project follows a systematic data analysis workflow.

Step 1 — Data Collection

The dataset was imported into Python using the Pandas library.

Step 2 — Data Exploration

The dataset was examined to understand:

- Number of rows and columns
- Data types
- Statistical summary
- Missing values
- Duplicate records

Step 3 — Data Cleaning

Several preprocessing techniques were applied:

- Handling missing values
- Filling categorical missing values
- Removing duplicate rows
- Converting date columns
- Creating additional date features

Step 4 — Exploratory Data Analysis

The cleaned dataset was analyzed using statistical methods and graphical visualization.

Data Cleaning Process

The following preprocessing operations were performed:

- Checked dataset dimensions
 - Identified null values
 - Filled missing values
 - Removed duplicate records
 - Converted date columns into datetime format
 - Created additional features from dates
 - Verified data types
 - Checked statistical summary
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Data Visualization

The following visualizations were created during analysis:

- Missing Value Analysis
- Histogram
- Box Plot
- Pie Chart
- Count Plot
- Correlation Heatmap
- Distribution Plot
- Top Categories Analysis
- Country-wise Analysis
- Release Year Analysis

These visualizations help identify trends, distributions, and relationships within the dataset.

Libraries Used

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Results

The analysis successfully demonstrated the complete data analysis workflow.

Key outcomes include:

- Successfully cleaned the dataset.
- Removed duplicate entries.
- Managed missing values efficiently.
- Created meaningful visualizations.
- Performed exploratory data analysis.
- Extracted valuable business insights from Netflix content.

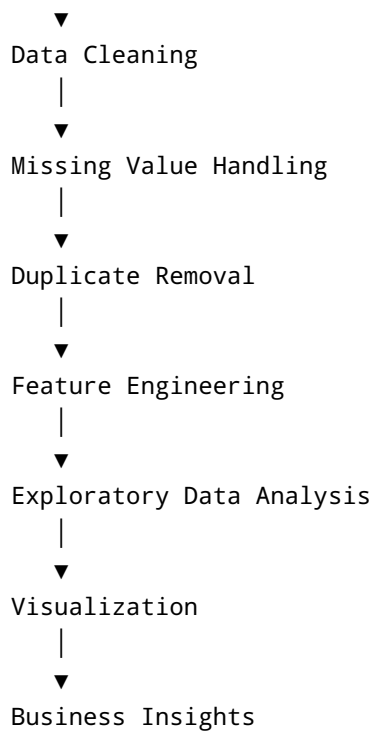
Business Insights

Some important observations from the analysis include:

- Netflix contains a larger number of Movies compared to TV Shows.
- Content production increased significantly in recent years.
- Certain countries contribute a major portion of Netflix's content library.
- Most content belongs to a limited number of popular genres.
- Ratings are concentrated among a few age classifications.
- The cleaned dataset is suitable for further machine learning and predictive analysis.

Project Workflow

```
Dataset
|
▼
Load Dataset
|
▼
Data Exploration
|
```



Conclusion

This project successfully demonstrates a complete data analysis workflow using Python. By applying data cleaning, preprocessing, exploratory analysis, and visualization techniques, meaningful insights were extracted from the Netflix dataset.

The project highlights the practical use of Pandas, NumPy, Matplotlib, and Seaborn in solving real-world data analysis problems. It also strengthens understanding of data preprocessing, statistical analysis, and visual storytelling, making it a strong portfolio project for students interested in Data Analytics, Data Science, and Machine Learning.

Future Scope

Future improvements may include:

- Interactive dashboards using Power BI or Tableau.
- Predictive analytics using Machine Learning.
- Recommendation systems.
- Sentiment analysis of movie descriptions.
- Time-series forecasting for content trends.

Skills Demonstrated

- Python Programming
 - Data Cleaning
 - Data Preprocessing
 - Exploratory Data Analysis (EDA)
 - Data Visualization
 - Statistical Analysis
 - Feature Engineering
 - Git & GitHub
 - Problem Solving
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References

1. Netflix Movies and TV Shows Dataset
 2. Pandas Documentation
 3. NumPy Documentation
 4. Matplotlib Documentation
 5. Seaborn Documentation
 6. Python Official Documentation
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Repository Structure

```
Netflix-Data-Analysis/  
|  
├─ Netflix_Data_Analysis.ipynb  
├─ netflix_dataset.csv  
├─ README.md  
├─ REPORT.md  
└─ images/  
    ├─ heatmap.png  
    ├─ histogram.png  
    ├─ pie_chart.png  
    ├─ countplot.png  
    └─ boxplot.png
```

Author

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Thank you for reviewing this project.